

Serpentine Laser Array

Localized Mode Switching on LED Panel (1x1 to 6x8 Expansion Study)

Red laser dot(s) projected onto a diffuse white LED panel surface, recording the spatial distribution and optical behavior of reflected/scattered light across 10 trials. Current configuration operates as a **1x1 serpentine array**, exhibiting localized mode switching rather than large-scale collective propagation. Session 2 (videos 07-10) confirms discrete satellite behavior in dark conditions and introduces **UV suppression** as a background control condition. Expansion to **6x8** is planned to achieve continuous fringe coupling and full lattice-level dynamics.

Video Captions — Session 1 (White Panel Background)

File	Dur.	Description
01_baseline_single_dot.mp4	~48s	Baseline — single localized mode. One stationary dot at lower-center. No lattice propagation; edge dominated.
02_dot_appears_bottom.mp4	~48s	Mode re-entry at edge. Dot reappears at bottom edge with slight positional drift. Early surface reflection visible.
03_dots_cluster_scatter.mp4	~47s	Discrete satellite cluster onset. Multiple dots in a loose, irregular cluster across the lower half. Transition from single to multiple.
04_dots_arc_pattern.mp4	~43s	Angular preference emerging. Dots arrange into a curved arc — non-circular symmetry indicates serpentine dynamics.
05_streaks_vertical.mp4	~35s	Continuous fringe to streak transition. Dots elongate into vertical streaks, suggesting partial fringe coupling.
06_streaks_full_panel.mp4	~34s	Broadest local mode extent. Multiple streaks across larger panel area — closest to collective behavior observed.

Video Captions — Session 2 (Dark Background / UV Suppression)

File	Dur.	Description
07_discrete_cluster_dark.mp4	~41s	Discrete satellite cluster confirmed — dark field. Panel background fully dark; red dots appear as isolated points.
08_cluster_to_streaks_dark.mp4	~33s	Cluster to streak transition in dark conditions. Satellite dots elongate into directional streaks as the trial progresses.
09_scattered_dots_dark.mp4	~38s	Wide-field discrete dot scatter — dark field. Dots distributed across a broader area in distinct, separated locations.
10_uv_suppression_red_on_blue.mp4	~36s	UV suppression trial. UV illumination introduced; panel background shifts to deep blue/violet. Red laser dot remains visible.

Technical Notes

- **Camera:** 1920x1080 @ ~30 fps (original)
- **Panel:** Diffuse white surface (LED panel / acrylic diffuser)
- **Laser:** Red (~650 nm), single-point source
- **Compressed:** 960x540, H.264 CRF 28, 15 fps — optimized for GitHub (<2 MB/file)

Analysis — Why It Feels Static (Current 1x1 Serpentine)

The following interpretation is based on GPT analysis of the current footage.

The current 1x1 serpentine configuration behaves more like **localized mode switching** than true dynamic rotation. The 'slightly static' impression is accurate, for the following reasons:

- No global lattice propagation occurs

- Coupling paths are short
- The edge effectively acts as the entire system

As a result, the pattern appears to **jump between discrete states** rather than flow continuously across the surface.

Expected Changes with 6x8 Expansion

- Fringe plumes will extend and connect spatially
- Satellite activations will propagate in sequence across the array
- The current 'frozen dot' appearance is expected to diminish significantly

Notable Finding — Angular Preference

The discrete satellite clusters in the later footage are **not fully circularly symmetric**. This suggests the serpentine geometry may be **inducing an angular preference** in the optical output. If this were simple diffuse scattering, rotating the source should only increase diffuse blur. Instead, **the same directional satellite set reappears consistently** — indicating that the structure is actively involved in **optical path selection**, not just passive scattering.

Summary Assessment (Updated — Sessions 1 & 2)

Item	Status
Continuous fringe to discrete satellite transition	Partially observed ✓
Behavior confined to local mode level	Confirmed ✓
Discrete character confirmed under dark field	Confirmed (Session 2) ✓
Red satellite survives UV background suppression	Confirmed (video 10) ✓
Large-array collective effect	Insufficient ✗

Conclusion: Discrete behavior is real and structurally driven, not an artifact of background conditions. However, the effect remains at local mode level. Expanding to a full **6x8 array** is necessary to achieve large-scale collective behavior.